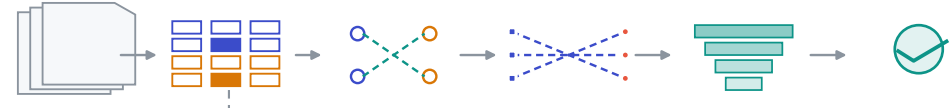functional branch